CampusSafe & Peer Evaluation — Master Development Progress Log

Course: COMP 018 — Database Administration
Project: CampusSafe: Campus Hazard & Safety Reporting System
Evaluating Group: Group 7 (Alba, Arreza, Orpesa, Robiso, Santos)
Repository: https://github.com/jeffersonorpesa2-ops/grp07-campussafe
Status: All Activities 1 to 6 & 8 Peer Evaluations Fully Complete (100/100)

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1. Team Member Roles & Focus Areas

Member Name	Assigned Focus	Key Primary Deliverables	Peer Evaluation Focus
Jefferson Orpesa (Lead)	Database Architecture & Administration	Schema DDL (`01_schema.sql`), Seed data (`02_seed.sql`), Table Constraints, Reset scripts, Activity 4 Roles Script	RBAC architecture design, privilege matrix formulation, and cross-database boundary lockdown
Alba	Backend Security & Transactions	Authentication (`Auth.php`), Session RBAC, CSRF defense, Transactional status transitions (`Report.php`)	Verification test suites (Test Suites 1, 2, and 3), security auditing, password hash shielding views
Arreza	Administration & Maintenance Modules	Verification queue (`verify.php`), Dispatch assignments (`assign.php`), Master data CRUD (`users.php`, `categories.php`, `locations.php`)	Table bloat inspection (`pg_stat_user_tables`), vacuuming/analyzing, and query optimization (`EXPLAIN ANALYZE BUFFERS`)
Robiso	Technician Portal & Reporting UI	Frontend UI design (`style.css`, `app.js`), Work order resolution (`update.php`), Search & filter (`list.php`), Attachments	Backup strategy execution (`pg_dump -Fc`), TOC verification, and SHA-256 checksum generation
Santos	Quality Assurance, Recovery & QA Testing	Documentation, System Architecture, Troubleshooting guide, Backup strategy (Activity 6), PDF progress reporting	Restoration verification (`pg_restore`), record parity audits, and Canary disaster recovery drills

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2. CampusSafe Internal Milestones (Activities 1 – 6)

Milestone	Phase	Description	Deliverables	Status
Activity 1	Requirements & Proposal	Campus safety domain definition, user stories, stakeholder requirements, and functional scope.	Project Proposal, Requirements Matrix	COMPLETED

Milestone	Phase	Description	Deliverables	Status
Activity 2	Relational Modeling & Schema	ER modeling, normalization to 3NF, table constraints, foreign key cascades, and check constraints.	ERD, Data Dictionary, `01_schema.sql`	COMPLETED
Activity 3	Application Implementation	Full MVC-lite PHP web portal with 10 functional modules, session auth, CSRF defense, and file uploads.	Full Web Application, 1-Click Setup, Seed Data	COMPLETED
Activity 4	Access Control & Security	Multi-tier RBAC (`app_runtime_grp`, `app_readonly_grp`, `app_backup_grp`), password hash masking, and isolation.	`03_roles_and_privileges.sql`, `db-access-control/`	COMPLETED
Activity 5	Maintenance & Query Tuning	Server health checks, table statistics, VACUUM/ANALYZE automation, and `EXPLAIN (ANALYZE, BUFFERS)` indexing.	`db-maintenance/`, Performance Audit	COMPLETED
Activity 6	Backup & Disaster Recovery	RPO 24h / RTO 2h backup strategy, custom-format binary dump, SHA-256 verification, and disaster simulation.	`db-backup-recovery/`, Canary Drill	COMPLETED

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3. Peer Evaluation Master Tracker (Activities 4, 5, and 6)

Group 7 conducted rigorous, hands-on, live-engine Database Administration evaluations for **8 classmate project teams** on PostgreSQL 18.6:

Target Group	Target Project Name	Target Repository	Scope Evaluated	Report Document Link	Evaluation Status
Group 4	BrewPOS: Coffee Shop Point-of-Sale System	BrewPOS Repository	Activities 4, 5, 6	`GROUP_04_BREWPOS_EVALUATION_REPORT.md`	**100/100 COMPLETED**
Group 2	Park-It: Smart Parking Lot Management System	Park-It Repository	Activities 4, 5, 6	`GROUP_02_PARKIT_EVALUATION_REPORT.md`	**100/100 COMPLETED**
Group 3	Project OJT RMA: Internship	OJT RMA Repository	Activities 4, 5, 6	`GROUP_03_OJT_RMA_EVALUATION_REPORT.md`	**100/100 COMPLETED**

Target Group	Target Project Name	Target Repository	Scope Evaluated	Report Document Link	Evaluation Status
	Management System				
Group 5	Lost & Found Management System	grp05_lostandfound	Activities 4, 5, 6	`GROUP_05_LOSTANDFOUND_EVALUATION_REPORT.md`	**100/100 COMPLETED**
Group 6	HabalGo: Motorcycle Ride-Hailing Platform	HabalGo Repository	Activities 4, 5, 6	`GROUP_06_HABALGO_EVALUATION_REPORT.md`	**100/100 COMPLETED**
Group 8	bwisIT Solutions: Campus Canteen Pre-Order System	Campus Canteen Repository	Activities 4, 5, 6	`GROUP_08_CANTEEN_EVALUATION_REPORT.md`	**100/100 COMPLETED**
Group 10	Binary Sisters: Restaurant Online Ordering & Table Reservation	Restaurant System	Activities 4, 5, 6	`GROUP_10_RESTAURANT_EVALUATION_REPORT.md`	**100/100 COMPLETED**
Group 11	BoardTrack: Boarding House & Rental Management System	BoardTrack Repository	Activities 4, 5, 6	`GROUP_11_BOARDTRACK_EVALUATION_REPORT.md`	**100/100 COMPLETED**

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4. Responsible AI Usage Statement (Activities 1 – 6)

Phase / Date	AI Tool Utilized	Specific Task Prompted	Human Validation & Modifications Made by Team	Verified By
Activity 1 & 2	Gemini / Antigravity	Relational schema constraint drafting and table generation	Verified PostgreSQL 14+ compatibility; added explicit `chk_verified_pair` and `chk_resolved_has_date` constraints; ensured cascade rules align with business requirements.	Jefferson Orpesa

Phase / Date	AI Tool Utilized	Specific Task Prompted	Human Validation & Modifications Made by Team	Verified By
Activity 3	Gemini / Antigravity	PDO transaction pattern boilerplate for status transitions & CSRF	Reviewed exception handling; added explicit row locking (`FOR UPDATE`) to prevent race conditions during concurrent admin reviews.	Alba
Activity 4	Antigravity AI Assistant	Role-Based Access Control matrix formulation, security view generation, and least-privilege testing scripts	Verified role segregation, verified that financial tables (`payment`, `order_item`, `lease`) have `DELETE` revoked; tested cross-database isolation against `grp07_campussafedb`.	Alba & Orpesa
Activity 5	Antigravity AI Assistant	`EXPLAIN (ANALYZE, BUFFERS)` execution plan extraction, dead tuple bloat calculation, and indexing strategy	Executed all `VACUUM` and `ANALYZE` commands live on PostgreSQL 18.6; verified 0 dead tuples post-vacuum; validated index query speedups.	Arreza
Activity 6	Antigravity AI Assistant	Backup automation script formatting, TOC parser, SHA-256 generation, and canary drill script structure	Ran live `pg_dump -Fc` and `pg_restore` into separate restore test databases; computed physical SHA-256 hashes; executed live canary disaster drills.	Robiso & Santos

AI Ethics Declaration: All AI-assisted code was manually inspected, line-by-line reviewed for SQL injection, authenticated role checks, cross-site scripting vulnerabilities, and relational schema integrity prior to team sign-off.

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5. Master Issues & Resolutions Table

#	Problem Encountered	Root Cause	Technical Resolution Applied
1	`Database Connection Error: could not find driver (SQLSTATE[08006])`	PHP CLI launched without an active `php.ini` file; PostgreSQL PDO drivers (`pdo_pgsql`, `pgsql`) were not loaded.	Created `D:\php\php.ini`, configured `extension_dir = "ext"`, enabled `extension=pdo_pgsql` and `extension=pgsql`, and restarted the server.
2	`Database Connection Error: FATAL: password authentication failed for user "postgres"`	Template `.env` password (`DB_PASSWORD=postgres`) mismatched local PostgreSQL master credentials.	Synchronized local environment configuration by updating `DB_PASSWORD` in `.env` to match the local workstation credentials.
3	Port conflict between PostgreSQL 14 and PostgreSQL 18 (Port 5432 vs. 5433)	Workstation had multiple PostgreSQL versions installed with different listening ports.	Used `Get-NetTCPConnection` to identify active port 5432, parameterized `DB_PORT` in `.env`, and updated connection DSN.
4	`Permission denied for sequence ..._seq` when	In PostgreSQL, granting `INSERT` on a table does not automatically	Added explicit sequence grant: `GRANT USAGE, SELECT ON ALL SEQUENCES IN SCHEMA public TO <group_role>;`.

#	Problem Encountered	Root Cause	Technical Resolution Applied
	inserting records using runtime account	grant `USAGE` / `SELECT` on its auto-increment sequence.	
5	Password hash exposure in read-only / auditor reporting queries	Granting direct `SELECT` on user credential tables (`app_user`, `customer`, `staff`, `accounts`) exposes bcrypt hashes.	Created sanitized security views (`v_user_directory`, `v_customer_directory`, `v_staff_directory`) that omit password hashes, granting `SELECT` on views while withholding direct table access.
6	Trigger-induced dead tuple bloat in tables with automated status sync	Inserting leases/orders triggers automated status updates on parent rooms/tables, producing high MVCC dead tuple bloat.	Scheduled routine automated `VACUUM (VERBOSE, ANALYZE)` operations, reducing dead tuples from 35+ down to exactly 0.
7	Accidental financial ledger or contract deletion via runtime account	Default full CRUD grants allow web application code to delete historical payments, lease agreements, or line items.	Enforced strict Least Privilege: granted `SELECT, INSERT, UPDATE` while strictly withholding `DELETE` on financial ledgers (`payment`, `order_item`, `lease`).
8	Sequence out-of-sync error after manual or seed ID insertions	Explicit primary key insertions during testing/seeding do not advance PostgreSQL serial sequences.	Executed `SELECT setval(pg_get_serial_sequence('...', '...'), (SELECT MAX(...) FROM ...));` post-seed and verified in post-restore suite.
9	Cross-database access breach in shared PostgreSQL cluster	Default `PUBLIC` role possessed `CONNECT` privilege on cluster databases, allowing accounts to connect to other groups' databases.	Executed `REVOKE CONNECT ON DATABASE <db> FROM PUBLIC;` and granted `CONNECT` exclusively to designated group roles.